* Case Study 1:

a) Bright Mart is a medium seized online shopping company, customers use its website and mobile application to purchase products.

The company store

i) Customers name
ii) Email addresses
iii) Phone numbers
iv) Delivery addresses
v) Passwords
vi) Order Information
vii) Payment-related information

BrightMart has around 100 employees.
The company recently hired several new employees. To make things easier, the IT department gave employees access to different internal systems!

However, the company has the following security weaknesses:
- Some employees use simple passwords such as BrightMart123.
- Many employees use the same password for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity training.
- There is no multi-factor authentication for some important accounts.
- Employees can access sensitive information that they do not necessarily need for their jobs.
- Backups are performed only once a month.

The incident

One monday morning, an employee named Rahul receives an email.
Subject: Urgent - Your Account will be Suspended

The email appears to come from the company's IT department.
It says:
"Your company account will be suspended today. Click the link below and verify your password immediately."
Rahul clicks the link.

The page looks almost identical to the company's login page.
He enters his username and password.
Rahul does not realize that the website is controlled by an attacker.
The attacker now has Rahul's login credentials.

Stage 1 - Unauthorized Access

The attacker uses Rahul's credentials to access Bright mart's internal system.
Because Rahul uses the same password for another system, the attacker is also able to access that system.
The ~~system~~ attacker discovers that Rahul has access to customer information.
~~The attacker discovers that Rahul has access to customer Satisfaction, inf~~ The attacker downloads a large amount of customer data.

1(g) What Cybersecurity problem has occured?

Stage2 - Malware Attack

The attacker sends another email to Rahul containing a malicious attachment.
Rahul opens the attachment.
Malware is installed on his computer.
The malware begins communicating with the attacker's system.
The attacker now attempts to move on from Rahul's computer to other systems within BM networks.

2g) Which Vulnerability allowed the attacker to move further into the organization?

Stage 3 - Ransomware

Two days later, employees arrive at work and discover that they cannot access important files.
A message appears on their computers:
"Your files have been encrypted. Pay the ransom to recover them".
The company's customer database and several important business files are ~~envy~~ encrypted.
unfortunately, the company's most recent backup is almost a month old.
BM website stops waiting.
Customers cannot place orders.
The customers - support team cannot access customer records.

Stage 4 - Business Impact

BM now faces several problems
Financial Impact

The company loses sales because its website is unavailable.

Data Impact

Customer information may have been stolen.

Operational Impact

Employees cannot access important systems.

Reputation Impact

Employees cannot access important system.

Legal / Compliance Impact

The company may face regulatory or legal consequences depending on what information was exposed and which laws apply.

Group 1 - Identify the Threats

(i) What threats are present in this case?

Group 2 - Identify the vulnerabilities

(i) What weaknesses in BM's Security allowed the attack to succeed?

Group 3 - Apply the CIA Traid

(i) Was confidential information exposed?
(ii) Was information changed or damaged?
(iii) Were systems available to legitimate users?

Group 4 - Security Solutions

(i) Who is responsible for cybersecurity?
(ii) Was phising the only problem?

(iii) What was the biggest vulnerability?

Solutions:-

Q1) A→ • Phishing / • Credential theft / • unauthorized access / • Data breach

Q2) A→ 1) weak passwords
2) Excessive user privileges
3) Lack of MFA
4) Poor network security
5) Lack of employee awareness
6) Outdated software

G1-Q) Threats present in this case?
A→ • Phishing • Malware • Ransomware • Credential theft • unauthorized access

G2-Q) What weakness in Bm's security allowed the attack to succeed?

A→

Vulnerability	How it helped the attacker
1) Weak Passwords	Easier to compromise accounts.
2) Password reuse	One Stolen password gives access to multiple systems.
3) No MFA	Stolen passwords may be enough to log in

4) Lack of training — Employee falls for phising

5) Delayed updates — Vulnerable software may be exploited.

6) Excessive Privileges — Attacker can access more data.

7) Infrequent backups — Difficult to recover after ransomware.

8) Group-4

A→
- use strong passwords
- Implement MFA
- Provide regular cybersecurity awareness training.
- Conduct phishing-awareness exercises.
- Keep software updated
- Follow the principle of least privilege.
- Segment important networks.
- Maintain regular backups.
- Encrypt sensitive data.
- Deploy antivirus/~~endpoint~~ endpoint protection.
- monitor suspicious login activity.
- Develop an incident-response plan.

C1-3 9) (i) A→ Yes. Customer information was accessed by the attacker.

(ii) Potentially yes. The attacker could modify or encrypt files.

(iii) No. The ~~web~~ website and internal systems became unavailable because of ransomware.

(4-8) i) A→ Not only the IT department. Every employee who uses the organization's systems has a role in cybersecurity.

ii) No. Phishing was the initial attack, but weaknesses such as passwords reuse, lack of MFA, excessive privileges, poor backups, and insufficient training allowed the attack to become much more damaging.

iii) There is no single correct answer.
1) Lack of employee awareness was a major vulnerability because the attack started when Rahul clicked the phishing link.

2) Poor backups pratices were the biggest problem because Bnt could not quickly recover from ransomware.

Both are valid.

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